Probability & Random Processes
EE2011

Department of Electrical Engineering

2.7
Bayes Rule

NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES, LAHORE CAMPUS

Conditional Probability

Definition 2.10:

The conditional probability of B, given A, denoted by $P(B|A)$, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided} \quad P(A) > 0.$$

Independent Events

Definition 2.11:

Two events A and B are **independent** if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A),$$

assuming the existences of the conditional probabilities. Otherwise, A and B are **dependent**.

The Product Rule, or the Multiplicative Rule

Theorem 2.10:

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

EXAMPLE :

If a coin is tossed two times then what is the probability of two Heads?

ANSWER :

$$\frac{1}{4} .$$

EXAMPLE :

If a coin is tossed two times then what is the probability of two Heads, *given that the first toss gave Heads* ?

ANSWER :

$$\frac{1}{2} .$$

Several examples will be about *playing cards* .

A standard *deck* of *playing cards* consists of 52 cards :

- Four *suits* :

  Hearts , Diamonds (*red*) , and Spades , Clubs (*black*)

- Each suit has 13 cards, whose *denomination* is

  $2\ ,\ 3\ ,\ \cdots\ ,10\ ,$ Jack , Queen , King , Ace .

- The Jack , Queen , and King are called *face cards* .

Theorem 2.11:

Two events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B).$$

Therefore, to obtain the probability that two independent events will both occur, we simply find the product of their individual probabilities.

The multiplicative rule

Theorem 2.10:

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

can be extended to more than two-event situations.

Theorem 2.12:

If, in an experiment, the events $A_1, A_2, \ldots, A_k$ can occur, then

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) \\ = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_k|A_1 \cap A_2 \cap \cdots \cap A_{k-1}).$$

If the events $A_1, A_2, \ldots, A_k$ are independent, then

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) = P(A_1)P(A_2) \cdots P(A_k).$$

Example 2.40:

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards.

Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs,

where A_1 is the event that the first card is a red ace,

A_2 is the event that the second card is a 10 or a jack,

and A_3 is the event that the third card is greater than 3 but less than 7.

***Solution*:** First we define the events

A_1: the first card is a red ace,

A_2: the second card is a 10 or a jack,

A_3: the third card is greater than 3 but less than 7.

Now $\quad P(A_1) = \frac{2}{52}, \quad P(A_2|A_1) = \frac{8}{51}, \quad P(A_3|A_1 \cap A_2) = \frac{12}{50},$

and hence, by Theorem 2.12,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$
$$= \left(\frac{2}{52}\right)\left(\frac{8}{51}\right)\left(\frac{12}{50}\right) = \frac{8}{5525}.$$

The property of independence stated in Theorem 2.11

Theorem 2.11:

Two events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B).$$

can be extended to deal with more than two events.

Consider, for example, the case of three events A, B, and C.

It is not sufficient to only have that $P(A \cap B \cap C) = P(A)P(B)P(C)$ as a definition of independence among the three.

Suppose $A = B$ and $C = \phi$, the null set.

Although $A \cap B \cap C = \phi$, which results in $P(A \cap B \cap C) = 0 = P(A)P(B)P(C)$, events A and B are not independent.

Hence, we have the following definition.

Definition 2.12:

A collection of events $\mathcal{A} = \{A_1, \ldots, A_n\}$ are mutually independent if for any subset of $\mathcal{A}$, $A_{i_1}, \ldots, A_{i_k}$, for $k \leq n$, we have

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k}).$$

2.7 Bayes' Rule

Bayesian statistics is a collection of tools

that is used in a special form of statistical inference

which applies in the analysis of experimental data in many practical situations in science and engineering.

Bayes' rule is one of the most important rules in probability theory.

Total Probability

Let an individual be selected at random from the adults of a small town to tour the country and publicize the advantages of establishing new industries in the town.

Table 2.1: Categorization of the Adults in a Small Town

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

Table 2.1: Categorization of the Adults in a Small Town

	Employed	**Unemployed**	**Total**
Male	460	40	500
Female	140	260	400
Total	600	300	900

Suppose that we are now given the additional information that 36 of those employed and 12 of those unemployed are members of the Rotary Club.

We wish to find the probability of the event A that the individual selected is a member of the Rotary Club.

Referring to Figure 2.12, we can write A as the union of the two mutually exclusive events $E \cap A$ and $E' \cap A$. Hence, $A = (E \cap A) \cup (E' \cap A)$,
and by Corollary 2.1 of Theorem 2.7, and then Theorem 2.10, we can write

$$P(A) = P[(E \cap A) \cup (E' \cap A)] = P(E \cap A) + P(E' \cap A)$$
$$= P(E)P(A|E) + P(E')P(A|E').$$

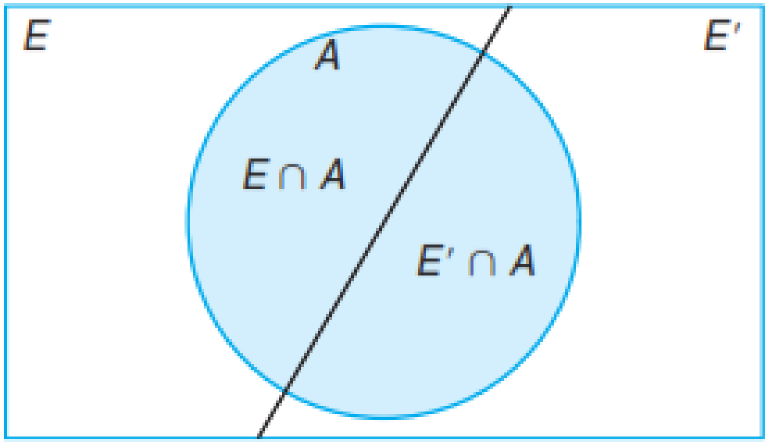


Figure 2.12: Venn diagram for the events A, E, and E'.

The data of Section 2.6, together with the additional data given above for the set A, enable us to compute

$$P(E) = \frac{600}{900} = \frac{2}{3}, \quad P(A|E) = \frac{36}{600} = \frac{3}{50},$$

$$\text{and} \quad P(E') = \frac{1}{3}, \quad P(A|E') = \frac{12}{300} = \frac{1}{25}.$$

If we display these probabilities by means of the tree diagram of Figure 2.13,

Table 2.1: Categorization of the Adults in a Small Town

	Employed	**Unemployed**	**Total**
Male	460	40	500
Female	140	260	400
Total	600	300	900

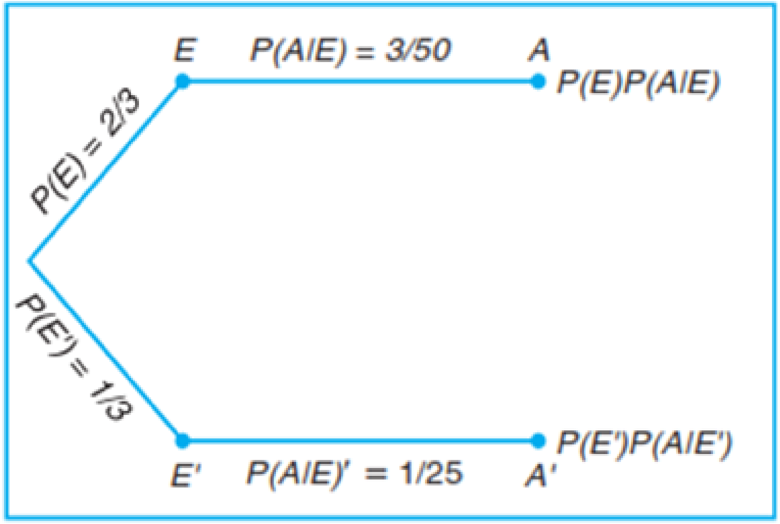
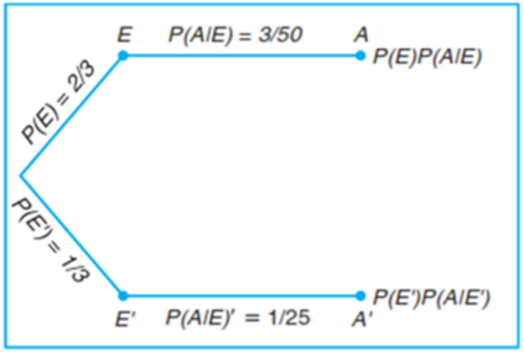


Figure 2.13: Tree diagram for the data on page 63, using additional information on page 72.

where the first branch yields the probability $P(E)P(A|E)$ and the second branch yields the probability $P(E')P(A|E')$,



it follows that $$P(A) = \left(\frac{2}{3}\right)\left(\frac{3}{50}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{25}\right) = \frac{4}{75}.$$

A generalization of the foregoing illustration to the case where the sample space is partitioned into k subsets is covered by the following theorem, sometimes called the **theorem of total probability** or the **rule of elimination**.

Theorem 2.13:

If the events $B_1, B_2, \ldots, B_k$ constitute a partition of the sample space S such that $P(B_i) \neq 0$ for $i = 1, 2, \ldots, k$, then for any event A of S,

$$P(A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i)P(A|B_i).$$

Proof: Consider the Venn diagram of Figure 2.14.

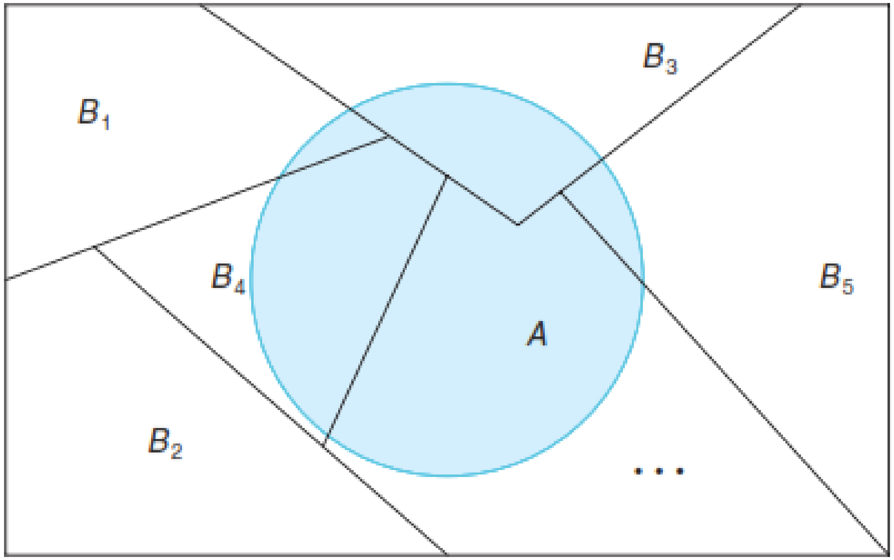


Figure 2.14: Partitioning the sample space S.

The event A is seen to be the union of the mutually exclusive events

$$B_1 \cap A,\ B_2 \cap A,\ \ldots,\ B_k \cap A;$$

that is,

$$A = (B_1 \cap A) \cup (B_2 \cap A) \cup \cdots \cup (B_k \cap A).$$

Using Corollary 2.2 of Theorem 2.7 and Theorem 2.10, we have

$$\begin{aligned}
P(A) &= P[(B_1 \cap A) \cup (B_2 \cap A) \cup \cdots \cup (B_k \cap A)] \\
&= P(B_1 \cap A) + P(B_2 \cap A) + \cdots + P(B_k \cap A) \\
&= \sum_{i=1}^{k} P(B_i \cap A) \\
&= \sum_{i=1}^{k} P(B_i)P(A|B_i).
\end{aligned}$$

Example 2.41:

In a certain assembly plant, three machines, B_1, B_2, and B_3, make 30%, 45%, and 25%, respectively, of the products.

. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective.

Now, suppose that a finished product is randomly selected.

What is the probability that it is defective?

Solution: Consider the following events:

A: the product is defective,

B_1: the product is made by machine B_1,

B_2: the product is made by machine B_2,

B_3: the product is made by machine B_3.

Applying the rule of elimination, we can write

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3).$$

Referring to the tree diagram of Figure 2.15, we find that the three branches give the probabilities

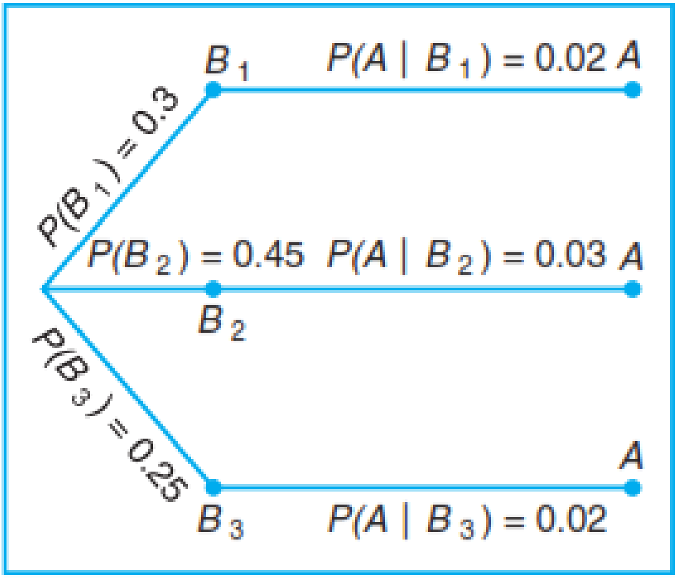


Figure 2.15: Tree diagram for Example 2.41.

$$P(B_1)P(A|B_1) = (0.3)(0.02) = 0.006,$$
$$P(B_2)P(A|B_2) = (0.45)(0.03) = 0.0135,$$
$$P(B_3)P(A|B_3) = (0.25)(0.02) = 0.005,$$

and hence

$$P(A) = 0.006 + 0.0135 + 0.005 = 0.0245.$$

Bayes' Rule

Instead of asking for $P(A)$ in Example 2.41, by the rule of elimination, suppose that we now consider the problem of finding the conditional probability $P(B_i|A)$.

In other words, suppose that a product was randomly selected and it is defective.

What is the probability that this product was made by machine B_i?

Questions of this type can be answered by using the following theorem, called **Bayes' rule**:

Theorem 2.14:

(Bayes' Rule) If the events $B_1, B_2, \ldots, B_k$ constitute a partition of the sample space S such that $P(B_i) \neq 0$ for $i = 1, 2, \ldots, k$, then for any event A in S such that $P(A) \neq 0$,

$$P(B_r|A) = \frac{P(B_r \cap A)}{\sum_{i=1}^{k} P(B_i \cap A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^{k} P(B_i)P(A|B_i)} \quad \text{for } r = 1, 2, \ldots, k.$$

Proof: By the definition of conditional probability,

$$P(B_r|A) = \frac{P(B_r \cap A)}{P(A)},$$

and then using Theorem 2.13 in the denominator, we have

$$P(B_r|A) = \frac{P(B_r \cap A)}{\sum_{i=1}^{k} P(B_i \cap A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^{k} P(B_i)P(A|B_i)},$$

which completes the proof.

Example 2.43:

A manufacturing firm employs three analytical plans for the design and development of a particular product.

For cost reasons, all three are used at varying times.

In fact, plans 1, 2, and 3 are used for 30%, 20%, and 50% of the products, respectively.

The defect rate is different for the three procedures as follows:

$$P(D|P_1) = 0.01, \qquad P(D|P_2) = 0.03, \qquad P(D|P_3) = 0.02,$$

where $P(D|P_j)$ is the probability of a defective product, given plan j.

If a random product was observed and found to be defective, which plan was most likely used and thus responsible?

Solution:

From the statement of the problem

$$P(P_1) = 0.30, \quad P(P_2) = 0.20, \quad \text{and} \quad P(P_3) = 0.50,$$

we must find $P(P_j|D)$ for $j = 1, 2, 3$.

Bayes' rule (Theorem 2.14) shows

$$P(P_1|D) = \frac{P(P_1)P(D|P_1)}{P(P_1)P(D|P_1) + P(P_2)P(D|P_2) + P(P_3)P(D|P_3)}$$

$$= \frac{(0.30)(0.01)}{(0.3)(0.01) + (0.20)(0.03) + (0.50)(0.02)} = \frac{0.003}{0.019} = 0.158.$$

Similarly,

$$P(P_2|D) = \frac{(0.03)(0.20)}{0.019} = 0.316 \text{ and } P(P_3|D) = \frac{(0.02)(0.50)}{0.019} = 0.526.$$

The conditional probability of a defect given plan 3 is the largest of the three; thus a defective for a random product is most likely the result of the use of plan 3. ┘